

Universitas Pandrosion: Paper 109

The de Bruijn–Newman Constant via Pandrosion–Pólya–Schur

A Pandrosion reformulation of the Newman conjecture $\Lambda \leq 0$, with honest scope limitations

I. Besevic

April 2026

Abstract

The de Bruijn–Newman constant Λ is defined via the heat-flow deformation $H_t(z) := \frac{1}{8} \int_0^\infty \Phi(u) \cos(zu) e^{tu^2} du$ where $\Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}$. H_0 corresponds (up to a substitution) to the Riemann ξ -function. The Riemann hypothesis is equivalent to H_0 having only real zeros.

Newman’s conjecture (1976): $\Lambda \leq 0$, i.e., H_t has only real zeros for all $t \leq 0$. This is implied by RH but is genuinely weaker.

Tao–Rodgers theorem (2018): $\Lambda \geq 0$.

Combined: $\Lambda = 0$ iff **RH holds at the boundary**. Newman conjectured $\Lambda = 0$, equivalent to RH being barely true.

This paper offers a Pandrosion reformulation of the Newman conjecture via the Pandrosion–Pólya–Schur framework (legacy paper 98) and the Pandrosion–Hadamard Gram matrix (legacy paper 77). The contribution is:

1. A reformulation: $\Lambda \leq t_0$ iff the Pandrosion field $F_{H_{t_0}}$ satisfies a Pólya–Schur stability condition.
2. Numerical exploration of H_t for small $|t|$, illustrating the regime near Λ .
3. An honest assessment: the Pandrosion machinery cannot close $\Lambda \leq 0$ (or the equivalent RH); we identify the missing input as a *Laguerre–Pólya class membership* of $\xi(s)$, which is precisely what is open.

We do not prove the Riemann hypothesis or any new bound on Λ .

Scope clarification. This is a reformulation paper. It does not advance the numerical bound on Λ (best known: $0 \leq \Lambda < 0.22$), nor does it prove RH or the Newman conjecture. It maps a sub-problem to the Pandrosion language.

Contents

1	Background	2
1.1	The Riemann ξ and H_0	2
1.2	Newman’s H_t and the de Bruijn–Newman constant	2
2	Pólya–Schur framework	2
3	Pandrosion–Hadamard reformulation	3
3.1	The Pandrosion field of polynomial truncations	3
3.2	Numerical exploration	3

4	Why Pandrosion does not close $\Lambda \leq 0$	3
4.1	The structural obstacle	3
4.2	What Pandrosion does provide	4
5	Honest conclusion	4

1 Background

1.1 The Riemann ξ and H_0

The Riemann ξ -function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

ξ is entire, satisfies $\xi(s) = \xi(1-s)$, and the Riemann hypothesis is equivalent to all zeros of ξ lying on the critical line $\Re(s) = 1/2$, i.e., $\xi(\frac{1}{2} + iz)$ has only real zeros in z .

1.2 Newman's H_t and the de Bruijn–Newman constant

Define $\Phi(u)$ as in the abstract. The function $H_t(z) := \frac{1}{8} \int_0^\infty \Phi(u) \cos(zu) e^{tu^2} du$ relates to ξ at $t = 0$ via $H_0(z) \propto \xi(\frac{1}{2} + iz)$.

The *de Bruijn–Newman constant* Λ is the unique value such that:

- For $t \geq \Lambda$, H_t has only real zeros.
- For $t < \Lambda$, H_t has at least one complex (non-real) zero.

Status.

- Newman (1976): conjectured $\Lambda \leq 0$.
- de Bruijn (1950s): $\Lambda \leq 1/2$.
- Csordas–Smith–Varga (1994): $\Lambda > -50$.
- Many subsequent improvements; current best: $0 \leq \Lambda < 0.22$ (Polymath15, 2018, building on Tao–Rodgers).
- Tao–Rodgers (2018): $\Lambda \geq 0$.
- Open: prove $\Lambda \leq 0$ (equivalent to RH at the boundary).

2 Pólya–Schur framework

The relevant operator-theoretic concept is the *Laguerre–Pólya class* $\mathcal{LP}$: entire functions of the form

$$f(z) = c z^m e^{-az^2+bz} \prod_k \left(1 - \frac{z}{\alpha_k}\right) e^{z/\alpha_k}, \quad a \geq 0, \alpha_k \in \mathbb{R}, \sum 1/\alpha_k^2 < \infty.$$

Theorem 2.1 (Pólya, classical). *$f \in \mathcal{LP}$ iff f is a uniform limit on compacta of polynomials with only real zeros.*

Pólya–Schur criterion (paper 98 of legacy corpus). A linear operator T on polynomials preserves real-rootedness iff its action on the basis $\{(z+1)^n\}_n$ is real-rooted (Pólya–Schur 1914). The Pandrosion–Pólya–Schur tools of paper 98 give algorithmic checks for $\mathcal{LP}$ -membership of polynomial truncations.

Connection to Λ .

Theorem 2.2 (Pólya equivalence). $\Lambda \leq t_0$ iff $H_{t_0} \in \mathcal{LP}$.

So Newman’s conjecture $\Lambda \leq 0$ becomes: $H_0 = \xi(\frac{1}{2} + i\cdot)$ is in $\mathcal{LP}$, equivalent to RH.

3 Pandrosion-Hadamard reformulation

3.1 The Pandrosion field of polynomial truncations

For each N , let $H_t^{(N)}(z) := \sum_{|n| \leq N} c_n(t) z^n$ be the Maclaurin truncation of H_t . Each $H_t^{(N)}$ is a polynomial; if its roots are real, then in the limit $N \rightarrow \infty$, $H_t \in \mathcal{LP}$ (under regularity hypotheses).

The Pandrosion field of $H_t^{(N)}$ is

$$F_{H_t^{(N)}}(z) = \frac{(H_t^{(N)})'(z)}{H_t^{(N)}(z)} = \sum_j \frac{1}{z - \alpha_j^{(N)}(t)},$$

where $\alpha_j^{(N)}(t)$ are the (complex) roots of $H_t^{(N)}$.

Proposition 3.1 (Pandrosion barrier for Λ). $\Lambda \leq t_0$ iff for every N sufficiently large, all roots $\alpha_j^{(N)}(t_0)$ are real, equivalently iff $\sum_j \text{Im}(1/(z - \alpha_j^{(N)}(t_0))) > 0$ for $z \in \mathbb{R}$.

This is the Pandrosion-Hurwitz criterion (legacy paper 67) applied to the polynomial sequence $\{H_t^{(N)}\}$.

3.2 Numerical exploration

The companion script `three_conjectures_attack.py` (Section “de Bruijn–Newman”) computes $H_t(z)$ via numerical quadrature of the integral $\frac{1}{8} \int_0^U \Phi(u) \cos(zu) e^{tu^2} du$ for $U = 4$ and a midpoint rule with 100 subintervals. For each t , it counts sign-changes of $H_t(z)$ along $z \in [-20, 20]$ as a proxy for the number of real zeros.

Remark 3.2 (Limitations of the truncation). *The cutoff $U = 4$ is far too small to resolve the genuine zeros of H_t at high frequencies (where $\xi(\frac{1}{2} + iz)$ has zeros near $z \approx 14.13, 21.02, 25.01, \dots$). A serious numerical investigation requires $U \rightarrow \infty$ via Riemann–Siegel-type asymptotics. The script reports only that H_t varies smoothly in t for small $|t|$, with no observable transition near $t = 0$ in the truncated regime.*

4 Why Pandrosion does not close $\Lambda \leq 0$

4.1 The structural obstacle

Pandrosion–Pólya–Schur (paper 98) gives algorithmic checks for real-rootedness of *specific* polynomials. To prove $\Lambda \leq 0$ (i.e., $H_0 \in \mathcal{LP}$), one would need:

1. Either show $H_0^{(N)}$ has only real zeros for every N (Hurwitz–Newton stability, paper 67) — this is essentially RH.
2. Or apply a stability-preserving operator that takes a known $\mathcal{LP}$ function to H_0 — no such operator is known.

The first route is RH itself. The second is the substance of the missing proof.

4.2 What Pandrosion does provide

The Pandrosion framework offers:

- **A unified operator language** via $F_{H_t} = H'_t/H_t$ (paper 76 index theorem, paper 89 Perron–Frobenius, paper 67 stability).
- **Numerical positivity tests** via $\operatorname{Im} F_{H_t}(z) > 0$ on $\mathbb{R}$ (paper 67 Schur stability).
- **Polynomial-truncation analysis** via the Pandrosion-Hadamard Gram matrix (paper 77): $\det G^{(H_t^{(N)})} = |\Delta(H_t^{(N)})|^2$ gives a discriminant-style invariant.

None of these translate the analytic difficulty of RH into a tractable Pandrosion question.

5 Honest conclusion

- **Established (this paper):** a Pandrosion-Pólya–Schur reformulation of $\Lambda \leq t_0$ as a stability condition on the Pandrosion field $F_{H_t^{(N)}}$.
- **Established (paper 98):** Pandrosion-Pólya–Schur tools for polynomial real-rootedness preservers.
- **Empirically explored:** the numerical behavior of H_t for small $|t|$ (severely truncated; no new bound on Λ).
- **Not established:** the Newman conjecture $\Lambda \leq 0$, the Riemann hypothesis, or any improvement on the current best bound $0 \leq \Lambda < 0.22$.

Open problem (unchanged). The Newman conjecture $\Lambda \leq 0$ remains open. The Pandrosion framework maps it to a Pólya–Schur stability statement on $\xi(\frac{1}{2} + i\cdot)$, but does not provide tools to prove it. This is consistent with the broader picture: RH-adjacent statements resist polynomial methods because ξ is genuinely transcendental.

What would Pandrosion contribute, if anything? A specific scenario where Pandrosion machinery could matter: if a stability-preserving operator $T \in \mathcal{T}_{\text{Polya} \text{---} \text{Schur}}$ applied to a known $\mathcal{LP}$ function (e.g., $\cosh$, $\sin$) produced H_0 , then $H_0 \in \mathcal{LP}$ (so $\Lambda \leq 0$, hence RH). No such T has been identified, in this paper or elsewhere.

References

- [1] C. M. Newman, *Fourier transforms with only real zeros*. Proc. Amer. Math. Soc. **61** (1976), 245–251.
- [2] N. G. de Bruijn, *The roots of trigonometric integrals*. Duke Math. J. **17** (1950), 197–226.
- [3] B. Rodgers, T. Tao, *The de Bruijn–Newman constant is non-negative*. Forum Math. Pi **8** (2020), e6.
- [4] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function*. arXiv:1809.10381 (2018).
- [5] G. Pólya, I. Schur, *Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen*. J. Reine Angew. Math. **144** (1914), 89–113.

- [6] I. Besevic, *Hadamard's Inequality and the Pandrosion Gram Matrix*. Pandrosion Dynamics **77** (2026).
- [7] I. Besevic, *Pólya–Schur and Pandrosion Stability Preservers*. Pandrosion Dynamics **98** (2026).
- [8] I. Besevic, *Stable and Hyperbolic Polynomials: A Unified Pandrosion Classification*. Pandrosion Dynamics **67** (2026).
- [9] I. Besevic, *The Polynomial Index Theorem via the Pandrosion Field*. Pandrosion Dynamics **76** (2026).